

Project P10 Research Notebook

Personalized Deep Learning Model for Closed-Loop 40 Hz Entrainment to Optimize
Theta-Gamma Phase-Amplitude Coupling in Alzheimer's Disease

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Section 1: Background Research and Project Selection

1.1 Disease Pathology and Gamma Oscillations

Alzheimer's disease (AD) affects over 55 million people worldwide and is projected to triple by 2050. The disease is characterized by two hallmark protein aggregations: extracellular amyloid-beta plaques and intracellular neurofibrillary tangles composed of hyperphosphorylated tau protein. These aggregations disrupt synaptic signaling, trigger neuroinflammation, and ultimately cause widespread neuronal death.

Gamma oscillations (30–100 Hz) are the highest-frequency brain rhythms and serve essential roles in memory encoding, sensory binding, attention, and long-range cortical communication. They are generated by the Pyramidal-Interneuron Network Gamma (PING) mechanism: fast-spiking parvalbumin-positive (PV) inhibitory interneurons fire at γ frequency and synchronize pyramidal neuron activity into coherent oscillatory patterns. In AD, γ power is reduced and cross-regional γ synchronization is impaired early in disease progression, in some cases before amyloid plaques are detectable. This disruption is not merely correlative but is understood to be a proximate cause of cognitive impairment.

Phase-amplitude coupling (PAC) quantifies how the amplitude of γ oscillations (38–42 Hz) is modulated by the phase of θ oscillations (4–8 Hz). Theta- γ PAC is the neural mechanism by which the brain organizes information for working memory: each θ cycle (125–250 ms) provides a temporal window in which γ bursts encode individual memory items. PAC is measured via the Modulation Index (MI), introduced by Tort et al. (2010), derived from the Kullback-Leibler divergence between the observed amplitude distribution across phase bins and a uniform distribution. MI ranges from 0 (no coupling) to 1 (perfect coupling). Multiple studies establish PAC as a sensitive biomarker for AD, with θ - γ coupling being the strongest predictor of working memory performance (standardized beta = 0.693, $p < 0.001$).

1.2 40 Hz Entrainment Therapy

In 2016, Iaccarino et al. (Nature) demonstrated that exposing AD model mice to 40 Hz flickering light triggered microglia to clear amyloid-beta plaques by 40–50% and upregulated genes associated with microglial phagocytosis. Subsequent research identified three therapeutic pathways: (1) microglial activation and phagocytosis via upregulated cytokines (IL-6, IL-4, M-CSF); (2) glymphatic clearance, where combined audiovisual 40 Hz stimulation promotes cerebrospinal fluid influx and interstitial fluid efflux; and (3) network restoration, enhancing default mode network (DMN) and frontoparietal connectivity (Lahijanian et al., 2024).

Clinical translation has progressed rapidly. Long-term 40 Hz auditory stimulation in rhesus monkeys (5+ weeks) elevated CSF amyloid-beta levels, indicating mobilization and clearance (PNAS, 2025). Cognito Therapeutics' Phase II human trial demonstrated slowed brain atrophy and preserved cognitive function, with plasma pTau217 reductions of 19–47% in responders (Chan et al., 2025). Martorell et al. (2019) showed that multi-sensory γ stimulation further reduced tau pathology and improved cognition in mouse models.

1.3 Literature Review Matrix

Study	Year	Model	Stimulation	Key Finding	Limitation
Iaccarino et al.	2016	AD mice (5XFAD)	40 Hz visual	40–50% amyloid reduction	Visual only; mice
Martorell et al.	2019	AD mice	40 Hz audiovisual	Reduced tau, improved cognition	Short-term sessions
Adaikkan et al.	2019	AD mice	40 Hz auditory	Microglial phagocytosis	No human data
Lahijanian et al.	2024	35 dementia patients	40 Hz auditory	Enhanced DMN connectivity	Fixed schedule only
Cognito Therapeutics	2025	Human Phase II	40 Hz audiovisual	Slowed atrophy, preserved cognition	30% non-responder rate

1.4 Research Gap Analysis

Current clinical protocols use fixed schedules: typically 40 seconds of stimulation followed by 20 seconds of rest, delivered identically for every patient. This approach has three fundamental problems:

1. **Wasted stimulation.** Approximately half of all stimulation is delivered when the brain is already well-entrained, providing no additional benefit while contributing to habituation.
2. **Missed therapeutic windows.** During rest periods, the brain frequently loses entrainment, but the fixed schedule cannot respond.
3. **Individual variability is ignored.** Clinical trial data shows approximately 30% of patients are non-responders. Among responders, trajectories diverge: some habituate (declining response), others sustain or increase their coupling.

No existing system predicts future entrainment state to enable proactive control. Current approaches either use fixed schedules (no adaptation) or reactive thresholds (respond after decline has already occurred, introducing delay). The concept of adaptive neuromodulation is well-established in other domains – closed-loop deep brain stimulation for Parkinson's reduces side effects and extends battery life – but has not been applied to 40 Hz γ entrainment therapy.

1.5 Research Questions and Hypotheses

Research question: Can a deep learning model predict future θ - γ PAC from real-time EEG, and can these predictions optimize 40 Hz auditory entrainment timing for Alzheimer's patients?

Hypotheses:

1. A causal temporal convolutional network can predict future PAC at 5–10 second horizons where simpler methods fail, enabling adaptive stimulation that targets periods of genuine therapeutic need.
2. Closed-loop predictive control will achieve higher alignment with optimal stimulation timing than fixed-schedule or reactive approaches.
3. The advantage of adaptive control will be consistent across individual patients and robust to different fatigue model assumptions.

1.6 Personal Motivation

My grandmother had dementia. I watched her lose the ability to recognize her own family. That experience motivated me to work on something that mattered. I have also played piano for eight years, and the intersection between auditory processing and brain rhythms drew me to neural entrainment research.

Section 2: Dataset Analysis and Technical Implementation

2.1 Dataset Characterization

Dataset: OpenNeuro ds005048 v1.0.1 – "40 Hz Auditory Entrainment in Dementia" (Lahijanian et al., 2024)

Parameter	Value
Subjects	35 elderly dementia patients
Location	Memory clinic, Tehran, Iran
EEG channels	19 monopolar (10/20 system)
Sampling rate	250 Hz
Stimulation	40 Hz amplitude-modulated auditory pulses (5 kHz carrier, 4% duty cycle)
Protocol design	Alternating 40-second stimulation / 20-second rest epochs
File format	BIDS-compliant .set files (MATLAB v7.3 HDF5) with companion .fdt files
Upstream preprocessing	Makoto's EEGLAB pipeline (1 Hz HP, 50 Hz notch, ICA, CAR)
Total download size	847 MB

Channel selection: 7 frontal channels were selected for analysis: Fp1, Fp2, F7, F3, Fz, F4, F8. Rationale: frontal regions show the strongest 40 Hz entrainment response and are the most accessible in clinical EEG setups. This reduces dimensionality from 19 to 7 channels without sacrificing entrainment-relevant information.

2.2 Data Loading Technical Challenges

Standard Python EEG tools (MNE-Python, EEGLAB readers) failed to load the MATLAB v7.3 HDF5 .set files. A custom loader was developed:

1. Read .set HDF5 metadata with h5py (top-level keys; no EEG wrapper group)
2. Read actual EEG data from companion .fdt files via `numpy.fromfile(dtype=float32)`
3. Reshape with `order='F'` (Fortran/column-major) – critical due to MATLAB storage convention
4. Extract stimulus/rest epoch boundaries from BIDS events.tsv

The Fortran-order reshaping was the most time-consuming bug to identify. MATLAB stores matrices in column-major order, while Python/NumPy default to row-major (C order). Without `order='F'`, the

data was silently transposed, producing invalid EEG signals that still passed basic sanity checks.

```
# Custom loader for MATLAB v7.3 .set files
with h5py.File(set_path, 'r') as f:
    n_channels = int(f['nbchan'][0, 0])
    n_samples = int(f['pnts'][0, 0])

# Read actual data from .fdt
data = np.fromfile(fdt_path, dtype=np.float32)
data = data.reshape((n_channels, n_samples), order='F')
```

2.3 Preprocessing Pipeline

The data was already preprocessed upstream by Makoto's EEGLAB pipeline. Light secondary preprocessing was applied:

Step	Method	Parameters
Bandpass filter	4th-order Butterworth, zero-phase (filtfilt)	0.5–80 Hz
Notch filter	Butterworth notch	50 Hz, Q = 30
Artifact rejection	Amplitude threshold	+/- 100 uV
Reference	Common average reference (CAR)	Applied after artifact rejection
Windowing	Sliding window	2.0 s (500 samples), 1.0 s hop (50% overlap)

2.4 PAC Computation: Modulation Index (Tort et al., 2010)

Frequency bands: Theta phase (4–8 Hz), γ amplitude (38–42 Hz).

Algorithm:

1. Bandpass filter the EEG into θ and γ using 4th-order Butterworth filters (zero-phase)
2. Apply Hilbert transform to extract instantaneous phase from θ and amplitude envelope from γ
3. Divide phase space $[-\pi, \pi]$ into 18 bins (20 degrees each)
4. Compute mean γ amplitude within each phase bin
5. Normalize to a probability distribution: $P_i = A_i / \sum(A)$
6. Compute KL divergence from uniform: $KL = \sum(P_i \log(P_i / n_bins))$
7. Normalize: $MI = KL / \log(n_bins)$

Critical methodological note: PAC was computed at the epoch level (full 20–40 second blocks) for measurement stability, then assigned uniformly to all constituent 2-second windows within that epoch. This means windows from the same epoch share identical PAC labels. This design choice ensures stable PAC estimates (short windows produce unreliable MI values) but limits the effective sample diversity for within-epoch prediction.

2.5 Subject-Level Data Split

Splits were performed at the subject level to prevent within-subject leakage between train/validation/test sets (random seed = 42):

Split	Subjects	Windows
Train	24	11,736
Validation	5	2,725
Test	6	2,822
Total	35	17,283

Output format: {train,val,test}_data.npz with windows shape (n, 1, 7, 500) and pac shape (n,).

PAC label statistics: range [0.000006, 0.000701], mean approximately 0.000044, standard deviation approximately 2.2×10^{-5} .

Section 3: Model Architecture Evolution

3.1 Overview: The Architecture Marathon

The central question was: can we predict instantaneous PAC from a single 2-second EEG window? Eight architectures were tested spanning three orders of magnitude in model size, from 200 parameters (Ridge regression) to 1.1 million (ViT-TCNet). The critical finding was that all architectures converged near $R^2 = 0.287$ on the test set, indicating a data ceiling rather than a model capacity limitation.

3.2 V1: EEGNet Baseline

Based on Lawhern et al. (2018), adapted for regression (original designed for classification).

Architecture:

Block	Layer	Details	Output Shape
Block 1	Conv2d (temporal)	8 filters, kernel (1,64), padding (0,32)	(B, 8, 7, 500)
	BatchNorm2d	8 features	(B, 8, 7, 500)
	Conv2d (depthwise spatial)	16 filters (D=2), kernel (7,1), groups=8	(B, 16, 1, 500)
	BatchNorm2d	16 features	(B, 16, 1, 500)
	ELU activation		(B, 16, 1, 500)
	AvgPool2d	kernel (1,4)	(B, 16, 1, 125)
	Dropout	p = 0.5	(B, 16, 1, 125)
Block 2	DepthwiseConv2d	16 filters, kernel (1,16), groups=16	(B, 16, 1, 125)
	PointwiseConv2d	16 filters, kernel (1,1)	(B, 16, 1, 125)

Block	Layer	Details	Output Shape
	BatchNorm2d, ELU, AvgPool2d(1,8), Dropout(0.5)		(B, 16, 1, 15)
Head	Flatten + Linear	240 -> 1	(B, 1)

Parameters: 1,457 total. **Test $R^2 = 0.287$.**

Training configuration:

- Loss: MSE on z-scored PAC targets (mean/std saved in checkpoint)
- Optimizer: Adam (lr = 0.001, weight_decay = 0.0001)
- Scheduler: ReduceLROnPlateau (factor = 0.5, patience = 5)
- Early stopping: patience = 15 epochs on validation loss
- Gradient clipping: max_norm = 1.0
- Batch size: 64

3.3 Feature Leakage Discovery

During V3 (SpecTempNet) development, PAC-derived features from `pac_features.py` were inadvertently included as model inputs. These features directly encode the target variable (the Modulation Index), creating a circular dependency. The model achieved $R^2 = 0.999$, which was immediately suspicious.

Detection methodology:

- Computed Pearson correlation between each feature and the PAC target
- Found correlations exceeding 0.7 for PAC-derived features
- Ridge regression coefficient analysis confirmed: PAC features carried 96.6% of total model weight
- Shuffle-label sanity check: model trained on permuted labels achieved $R^2 = -0.332$, confirming that the original signal was real (not an artifact of feature engineering)

Impact: All subsequent models excluded PAC-derived features from the input pipeline for static prediction. A dedicated audit script (`audit_leakage.py`) was created to flag any future reintroduction of leaky features. This experience was one of the most important lessons of the project: feature leakage is insidious and can produce results that look valid until scrutinized.

3.4 Architecture Exploration Summary

Version	Architecture	Parameters	Test R^2	Outcome
V1	EEGNet	1,457	0.287	Baseline; became final static model
V2	EEGNetV2 (delta-PAC)	3,200	0.06	Delta-PAC prediction failed
V3	SpecTempNet	180,000	0.236 (clean)	Originally 0.69 (leaked); cleaned R^2 lower

Version	Architecture	Parameters	Test R^2	Outcome
V4	ViT-TCNet	1,100,000	0.252	Overfitting; 1.1M params on 11K samples
V5	Ridge Regression	~200	0.287	Matched EEGNet with simplest model
V6	Optimized Ensemble	~500	0.287	No improvement over Ridge alone
V7	Raw EEG CNN+Attention	~50,000	0.28	No advantage over handcrafted features
V8	ATCNet/TransformE EG	~25,000	0.22	Underperformed Ridge

Key insight: The simplest models (EEGNet with 1,457 parameters, Ridge with ~200 parameters) matched the best performance of the most complex (ViT-TCNet with 1.1 million parameters performed worse). $R^2 = 0.287$ represents a data ceiling: epoch-level PAC labels assigned to 2-second windows inherently limit prediction accuracy. The signal-to-noise ratio was computed as -4.73 dB, confirming that the underlying signal is weak relative to noise.

Architecture capacity analysis: The ratio of training samples (11,736) to parameters is critical. EEGNet (8.1 samples/parameter) and Ridge (58.7 samples/parameter) fell in the optimal range. ViT-TCNet (0.01 samples/parameter) was severely overparameterized.

3.5 Decision: Why EEGNet Was Selected

EEGNet was selected for the closed-loop pipeline because:

1. It matches the best R^2 (0.287) achieved by any architecture
2. It has the fewest parameters (1,457) among deep learning models, enabling real-time inference on embedded devices
3. Inference time is under 1 millisecond on Apple Silicon (MPS), well within the 1-second decision cycle
4. The architecture is well-validated in the BCI literature (Lawhern et al., 2018)

Section 4: Temporal Prediction Innovation

4.1 Reframing the Problem

Since no single-window architecture could exceed $R^2 = 0.287$ for instantaneous PAC prediction, the approach was reframed: instead of predicting PAC better from one snapshot, predict it further into the future from a sequence of snapshots. This enables proactive rather than reactive control.

The critical question became: at what prediction horizon does the TCN provide useful predictions where simpler methods fail?

4.2 Feature Engineering for Temporal Sequences

Each 2-second window produces a 73-dimensional feature vector, organized into three categories:

Spectral Features (61 dimensions):

- Band power in 5 frequency bands: delta (0.5–4 Hz), θ (4–8 Hz), alpha (8–12 Hz), beta (12–30 Hz), γ (30–42 Hz)
- Computed independently for each of 7 channels: $5 \times 7 = 35$ features
- Cross-channel spectral coherence measures: 26 additional features
- Extracted via FFT-based power spectral density estimation

PAC-Derived Features (7 dimensions):

- `pac_current`: Current epoch-level PAC value
- `pac_ma2`, `pac_ma4`, `pac_ma8`, `pac_ma16`: Causal moving averages over 2, 4, 8, and 16 timesteps
- `pac_diff1`: PAC difference from $t-1$ to t (first derivative)
- `pac_diff4`: PAC difference from $t-4$ to t (longer-range trend)

All moving averages are strictly causal (use only past and current values). These features are safe for temporal prediction because the model predicts future PAC (at time $t+h$), not current PAC. The current PAC value serves as a legitimate input for forecasting.

Stimulation Context Features (5 dimensions):

- `stim_state`: Binary indicator (0 = rest, 1 = stimulation), from BIDS events.tsv
- `time_since_switch_60s`: Seconds since last stim/rest transition, clipped to [0, 1]
- `stim_frac_20s`: Fraction of time in stimulation over the preceding 20 seconds
- `cycle_phase_sin`: $\sin(2\pi t/60)$ encoding position within the 60-second stim/rest cycle
- `cycle_phase_cos`: $\cos(2\pi t/60)$ (complementary sinusoidal encoding)

4.3 Sequence Construction

Sequences were built per-subject to prevent cross-subject contamination:

- Lookback window: $T = 20$ timesteps (20 seconds of history at 1 Hz)
- Prediction horizon: $h = 5$ timesteps (5 seconds ahead)
- Target: raw PAC at $t+h$ (`target_smooth_window = 1`, no smoothing)
- Delta target: $y_{\text{future}} - y_{\text{current}}$ (auxiliary task)
- Feature normalization: z-score computed from training data only, applied to all splits

Sample counts (horizon = 5, ts = 1):

- Train sequences: 11,160
- Validation sequences: 2,605
- Test sequences: 2,678

4.4 Target Smoothing Investigation

An important methodological finding: target smoothing dramatically affects evaluation metrics and must be controlled when comparing models.

Target Definition	Test R ²	RMSE	Interpretation
Smoothed (ts = 5)	0.74	1.8 ×10	Predicting denoised PAC state
Raw (ts = 1)	0.17	3.3 ×10	Predicting actual noisy PAC

Decision: Raw PAC (ts = 1) was used for all primary results. Real-world control systems must handle noisy signals, not idealized smooth targets. Using ts = 5 inflates R² by making the target more autocorrelated and easier to predict, but overestimates clinical utility. This is a critical methodological consideration: models should never be compared across different target definitions.

4.5 MultiscaleCausalTCN Architecture

Input: (batch, 20, 73) – 20 timesteps, 73 features per step.

Component	Layer Details	Parameters
Input projection	Linear(73 -> 64) + LayerNorm(64) + SiLU	4,864
TCN Block 1 (d=1)	Causal DepthwiseConv1d(64, k=3, d=1, g=64) + PointwiseConv1d(64->64) + GroupNorm(1,64) + SiLU + Dropout(0.1) + Residual	4,608
TCN Block 2 (d=2)	Same structure, dilation=2	4,608
TCN Block 3 (d=4)	Same structure, dilation=4	4,608
TCN Block 4 (d=8)	Same structure, dilation=8	4,608
Attention pooling	Conv1d(64->1, k=1) + softmax over time -> weighted sum	65
Future head	Linear(64->64) + SiLU + Dropout + Linear(64->1)	4,225
Delta head	Linear(64->64) + SiLU + Dropout + Linear(64->1)	4,225
Total		31,043

Key design decisions:

- Causal padding:** Each depthwise convolution pads only on the left: $\text{F.pad}(x, ((\text{kernel_size}-1)*\text{dilation}, 0))$. The model cannot access future timesteps, which is essential for a real-time prediction system. This was verified by the audit pipeline.
- Dilation pattern [1, 2, 4, 8]:** Receptive field = $1 + (3-1) \times (1+2+4+8) = 31$ timesteps, covering the full 20-step input sequence with margin.
- GroupNorm(1, C) instead of BatchNorm:** Equivalent to instance normalization over channels. Chosen for stability with small batch sizes and cross-subject distribution shifts. Different patients have different PAC baselines; GroupNorm is invariant to batch composition.

4. **Depthwise-separable convolutions:** Factorize spatial and channel mixing for parameter efficiency. The full TCN uses only 31,043 parameters vs hundreds of thousands for standard convolutions with equivalent receptive field.

5. **Attention pooling:** Learned softmax weights over the time axis allow the model to focus on the most informative timesteps rather than using only the last timestep or a simple average.

6. **Dual-head output:** Future PAC prediction (primary) and delta-PAC prediction (auxiliary). The delta head provides a multi-task learning signal during training, though the best checkpoint used $\lambda_{\text{delta}} = 0.0$.

4.6 Training Configuration

Parameter	Value
Loss	Huber ($\delta = 1.0$) on future PAC
Optimizer	AdamW ($\text{lr} = 0.001$, $\text{weight_decay} = 0.001$)
Scheduler	ReduceLROnPlateau ($\text{factor} = 0.5$, $\text{patience} = 5$, monitoring val R^2)
Early stopping	Patience = 20 epochs on validation R^2
Gradient clipping	$\text{max_norm} = 1.0$
Batch size	128
Best epoch	53
Multi-task penalties	$\lambda_{\text{delta}} = 0.0$, $\lambda_{\text{consistency}} = 0.0$

Huber loss vs MSE: Huber loss ($\delta = 1.0$) was chosen over MSE for robustness to PAC outliers. PAC values span two orders of magnitude, and a few extreme values disproportionately influence MSE. Huber loss transitions from quadratic (near zero error) to linear (large error), reducing the impact of outliers on gradient updates.

4.7 TCN Performance (Horizon = 5, $t_s = 1$)

Metric	Value
Validation R^2	0.411
Test R^2	0.170
Test RMSE	3.3×10
Test Pearson r	0.433
Best validation R^2	0.411

The validation-test gap (0.411 vs 0.170) reflects the challenge of cross-subject generalization. The 5 validation subjects happened to have more predictable PAC dynamics than the 6 test subjects. With only 5–6 subjects per split, individual differences dominate split-level metrics.

Section 5: Horizon Sweep Analysis

5.1 Methodology

This is the central experiment of the project. Separate models were trained to predict PAC at horizons of 1, 2, 3, 5, 8, and 10 seconds. At each horizon, the TCN was compared against two baselines:

1. **Persistence baseline:** Predict that future PAC equals the current PAC value ($y_{\text{hat}} = y_t$). This is a strong baseline when PAC changes slowly.
2. **Ridge regression:** Linear model on the same 73-feature input, flattened across timesteps. Tests whether the TCN's advantage comes from its architecture or simply from having access to the right features.

All models in the sweep used `target_smooth_window = 5` to ensure fair comparison (all models predict the same target). The sweep demonstrates relative advantage across horizons, not absolute prediction accuracy.

5.2 Complete Horizon Results

Horizon	TCN R^2	Persistence R^2	Ridge R^2	TCN Margin over Persistence
1s	0.735	0.760	0.812	-0.025 (baselines win)
2s	0.470	0.488	0.542	-0.018 (baselines win)
3s	0.277	0.234	0.253	+0.043 (TCN wins)
5s	0.254	-0.267	-0.393	+0.521 (TCN dominates)
8s	0.240	-0.276	-0.211	+0.516 (TCN dominates)
10s	0.278	-0.256	-0.212	+0.534 (TCN dominates)

Source: `models/sweep_horizons_results.json` (verified against all reported values).

5.3 Interpretation

At 1–2 second horizons, PAC changes slowly enough that simply repeating the last observed value is a strong predictor (persistence $R^2 = 0.76$ at 1s). Ridge regression, which can combine multiple features linearly, performs even better ($R^2 = 0.81$ at 1s). The TCN adds no value at these short horizons.

At 3 seconds, the crossover point occurs. The TCN's R^2 (0.277) exceeds both persistence (0.234) and Ridge (0.253). The margin is modest (+0.04) but meaningful.

At 5–10 seconds, both baselines collapse to negative R^2 (worse than predicting the population mean). Persistence drops to -0.27 and Ridge to -0.39 at 5 seconds. The TCN maintains R^2 of 0.24–0.28 across this entire range, yielding a +0.5 R^2 margin. This margin is the TCN's value proposition: it is the only model that provides useful predictions at the timescale where a controller needs them for proactive decision-making.

The 5–10 second range is operationally relevant because a closed-loop controller needs at least 3–5 seconds of lead time to make proactive decisions (process the prediction, decide on action, ramp stimulation up or down smoothly).

5.4 Why Baselines Fail at Long Horizons

PAC computed from 2-second windows has near-zero temporal autocorrelation beyond a few seconds. The persistence baseline assumes $y_{t+h} = y_t$, which works when h is small relative to the autocorrelation timescale. When h exceeds this timescale, persistence predictions become uncorrelated with actual values, yielding $R^2 < 0$ (higher variance than the mean predictor). Ridge regression suffers the same fundamental limitation: linear extrapolation from current features cannot capture the nonlinear temporal dynamics that govern PAC evolution over 5–10 seconds.

The TCN succeeds because its dilated causal convolutions with receptive field of 31 timesteps can learn temporal patterns across multiple timescales (1, 2, 4, and 8 seconds via dilation), capturing longer-range dependencies that linear models miss.

Section 6: Closed-Loop Controller Integration

6.1 System Architecture

The complete closed-loop system operates in two stages:

Stage 1 – Static PAC Estimation (EEGNet):

- Input: Raw 2-second EEG window (7 channels, 500 samples)
- Output: Current PAC estimate
- Latency: < 1 ms on Apple Silicon

Stage 2 – Temporal PAC Forecasting (TCN):

- Input: 20 timesteps of 73-dimensional features (including Stage 1 PAC estimates)
- Output: Predicted PAC 5 seconds ahead + predicted delta-PAC
- Latency: < 2 ms on Apple Silicon

Controller Decision Pipeline: EEG window -> EEGNet -> PAC estimate -> Feature extraction (73 features) -> TCN -> Predicted future PAC -> PersonalizationModule (30s rolling baseline) -> z-score -> Decision logic -> STIMULATE / REST / MAINTAIN

6.2 PersonalizationModule

Maintains a rolling circular buffer of the 30 most recent PAC values (30 seconds at 1 Hz decision rate):

$$z = (\text{PAC}_{\text{current}} - \text{mean}(\text{buffer})) / (\text{std}(\text{buffer}) + 1\text{e-}8)$$

Requires a minimum of 10 samples before producing z-scores. This personalization ensures that thresholds adapt to each patient's baseline PAC level, which varies by an order of magnitude across the cohort.

6.3 Decision Logic

Reactive Threshold Controller:

Condition	Action
$z < -0.5$	STIMULATE (PAC below baseline, coupling is weak)
$z > +0.5$	REST (PAC above baseline, coupling is strong)
$-0.5 \leq z \leq +0.5$	MAINTAIN current state

TCN Predictive Controller: Extends the reactive controller with lookahead:

1. Feed 20 timesteps of features into TCN
2. If predicted delta-PAC < -0.3 : STIMULATE proactively (PAC predicted to decline)
3. If predicted delta-PAC $> +0.3$: REST proactively (PAC predicted to rise)
4. Otherwise: fall back to reactive z-score logic

Hysteresis: Minimum 5 seconds in any state before transitioning. This prevents rapid oscillation between stimulate and rest states, which would be clinically undesirable and potentially harmful.

6.4 Validation Methodology

All 35 subjects' EEG recordings were replayed through each controller in an offline counterfactual analysis. At each 2-second window, the controller made a stimulate/rest decision based on its strategy. The decision was compared against the true PAC state to compute alignment metrics. For validation, ground-truth PAC labels were used as TCN input, isolating the TCN's predictive contribution from EEGNet estimation error.

Controllers compared:

1. Fixed Schedule (clinical standard): 40s ON / 20s OFF
2. Reactive Threshold: z-score decisions on current PAC
3. TCN Predictive: z-score + 5-second lookahead
4. Hybrid TCN+Reactive: combines both signals with weighted ensemble
5. PI Controller: proportional-integral control on PAC error
6. Alignment Oracle: perfect knowledge of future PAC (theoretical upper bound)

6.5 Controller Comparison Results (N = 35 Subjects, Real EEG)

Controller	Alignment	Low-PAC Stim	High-PAC Rest	Stim %	PAC Gap (x10)
Fixed Schedule	45.0%	61.4%	28.6%	66.6%	-6.6
Reactive Threshold	64.5%	51.7%	77.3%	36.7%	+21.1
TCN Predictive	72.1%	82.6%	61.6%	59.7%	+30.5
Hybrid TCN+Reactive	73.8%	85.3%	62.2%	60.8%	+34.0
PI Controller	66.1%	38.6%	93.6%	22.0%	+27.2
Alignment Oracle	100.0%	100.0%	100.0%	48.3%	+33.3

Source: results/tcn_validation_results.json (all values verified).

Metric definitions:

- **Alignment:** (Low-PAC Stim Rate + High-PAC Rest Rate) / 2. Balanced accuracy of targeting. 50% = random, 100% = perfect.
- **Low-PAC Stim Rate (sensitivity):** Fraction of below-median PAC windows where controller stimulates. "Did we treat when the brain needed it?"
- **High-PAC Rest Rate (specificity):** Fraction of above-median PAC windows where controller rests.
- **PAC Gap:** Mean PAC during rest minus mean PAC during stim. Positive = controller correctly concentrates stimulation during low-PAC periods.
- Low/high split at each subject's median PAC (personalized threshold).

6.6 Key Observations

1. **Fixed Schedule stimulates in the wrong direction.** Its PAC gap is -6.6×10 (stimulating more during high-PAC than low-PAC periods), because its timing is uncorrelated with brain state.
2. **TCN achieves 92% of oracle performance** on PAC targeting gap ($30.5 / 33.3 = 91.6\%$, rounded to 92%).
3. **TCN targets 82.6% of low-PAC windows** for stimulation, compared to Reactive's 51.7%. This 60% improvement in therapeutic precision means the controller concentrates stimulation where it is most needed.
4. **Trade-off:** Reactive achieves higher high-PAC rest rate (77.3% vs 61.6%) because it is more conservative, using only 36.7% stimulation. However, this conservatism means it misses nearly half of the low-PAC windows that need treatment.
5. **TCN uses less stimulation than Fixed Schedule** (59.7% vs 66.6%) with dramatically superior targeting.

6.7 Statistical Significance (TCN vs Reactive)

All comparisons use Wilcoxon signed-rank tests (non-parametric, paired) with Hedges' g effect sizes and bootstrap 95% confidence intervals.

Metric	TCN	Reactive	Hedges' g	95% CI	p-value
Alignment	72.1%	64.5%	+1.31	[+0.75, +1.87]	< 0.001
Low-PAC Stim	82.6%	51.7%	+4.47	[+3.32, +5.62]	< 0.001
High-PAC Rest	61.6%	77.3%	-2.41	[-3.14, -1.68]	< 0.001
PAC Gap	+30.5	+21.1	+1.57	[+0.98, +2.17]	< 0.001
Lead Time (s)	0.79	0.20	+0.76	[+0.25, +1.26]	0.0003
Clinical Utility	0.681	0.591	+0.95	[+0.43, +1.47]	< 0.001

All primary metrics show large effect sizes ($g > 0.8$). The low-PAC targeting rate shows a very large effect ($g = 4.47$), reflecting the TCN's dramatically superior ability to concentrate stimulation during therapeutic windows.

Per-subject consistency: 35 out of 35 subjects show higher alignment with the TCN controller than with reactive control (binomial test, $p < 0.001$). The advantage holds across training (24 subjects), validation (5 subjects), and test (6 subjects) splits.

Section 7: Advanced Analysis and Validation

7.1 Habituation Analysis (Real Data)

A key motivation for adaptive control is neural habituation: the brain progressively "tunes out" repetitive stimulation. PAC trajectories were analyzed across stimulation blocks for all 35 subjects.

Population-level results:

Metric	Value
First block mean PAC	0.000996
Last block mean PAC	0.001040
Mean change	+4.5% (not significant)
Paired t-test	$t(34) = -0.616$, $p = 0.542$
Subjects declining (habituating)	17/35 (48.6%)
Subjects increasing (facilitating)	18/35 (51.4%)
Within-block habituation	46.7% of blocks show decline

Individual variability: Subject-level changes ranged from -66.8% (strong habituation) to +149.1% (strong facilitation). This heterogeneity is the fundamental finding: there is no population-level trend because the cohort splits nearly equally between habituators and facilitators. A fixed-schedule protocol optimized for the "average" patient is suboptimal for almost everyone. This validates the core premise of personalized, adaptive control.

7.2 Fatigue Sensitivity Analysis (Simulation)

Adaptive vs. fixed scheduling was tested across six fatigue severity levels using the EntrainmentSimulator and FatigueAwareSimulator, with 10 trials per condition and 360-second simulations:

Fatigue Rate	Fixed Efficiency	Adaptive Efficiency	Gain (%)	p-value
0.000 (none)	5.381	5.401	+0.39%	0.492 (n.s.)
0.004 (mild)	5.294	5.361	+1.27%	0.020
0.008 (moderate)	5.217	5.329	+2.14%	0.010
0.015 (mod-high)	5.101	5.231	+2.55%	0.010
0.025 (high)	4.968	5.184	+4.35%	0.002
0.040 (severe)	4.819	5.092	+5.65%	0.010

Source: results/fatigue_sensitivity.json (n = 10 trials per condition).

The adaptive advantage is non-significant without fatigue ($p = 0.492$) but grows monotonically with fatigue severity, reaching +5.65% under severe habituation. This pattern confirms that adaptive scheduling becomes increasingly valuable as habituation worsens, precisely when it matters most clinically.

7.3 Fatigue Model Robustness

Four different mathematical models of neural fatigue were tested to ensure results are not artifacts of a specific habituation assumption:

Fatigue Model	Improvement	Hedges' g	p-value
Exponential Decay	+9.0%	2.31	< 0.001
Step Function	+6.9%	1.21	< 0.001
Heterogeneous (50/50)	+8.9%	1.71	< 0.001
Saturation (synaptic)	+19.0%	3.66	< 0.001

All $p < 0.001$ across all fatigue model assumptions. The advantage holds regardless of how habituation is modeled, with the saturation model (biologically most plausible, modeling synaptic resource depletion) showing the largest benefit.

7.4 Threshold Robustness Analysis

The TCN controller's delta-z threshold was swept from 0.1 to 1.0 to verify that results are not artifacts of threshold tuning:

Delta-z	Alignment	Low-PAC Stim	PAC Gap (x10)
0.1	59.6%	51.2%	12.4
0.2	68.5%	72.9%	26.3
0.3	73.7%	84.9%	32.4
0.4	73.9%	85.3%	33.7
0.5	73.7%	85.3%	33.8
0.8	73.8%	85.3%	34.0
1.0	73.8%	85.3%	34.0
<i>Reactive baseline</i>	<i>64.5%</i>	<i>51.7%</i>	<i>21.1</i>

Source: results/threshold_sweep.json.

Performance plateaus at delta-z ≥ 0.3 , and the TCN outperforms the reactive baseline at all thresholds ≥ 0.2 . This stability indicates the result is robust to threshold selection and not dependent on specific parameter tuning.

7.5 Data Integrity Verification

Check	Result
Subject leakage between splits	PASS – no overlap verified
Temporal causality in features	PASS – all feature indices < target indices
Normalization leakage	PASS – scalars fit on training data only
Shuffle-label sanity	$R^2 = -0.332$ (confirms real signal, not artifacts)
Persistence baseline	Matches expected values at all horizons
Random seed reproducibility	All seeds fixed (seed = 42)
Feature-target correlation audit	PASS – no features exceed $r = 0.5$ with target

Section 8: Statistical Rigor and Reproducibility

8.1 Cross-Validation Strategy

Data was split at the subject level (no within-subject leakage): 24 train, 5 validation, 6 test subjects. This ensures that model performance reflects generalization to unseen patients, not memorization of individual EEG patterns.

Temporal split considerations: Within each subject's recording, temporal ordering was preserved in sequence construction. Sequences were built per-subject with no shuffling across subjects. The lookback window ensures that each input sequence comes from a continuous temporal segment.

Limitation: With only 35 subjects, the validation and test sets contain 5 and 6 subjects respectively. Individual subject variability dominates split-level metrics, which explains the validation-test R^2 gap (0.411 vs 0.170).

8.2 Effect Size Interpretation

All effect sizes reported use Hedges' g (bias-corrected standardized mean difference) with 95% bootstrap confidence intervals. Interpretation follows Cohen's guidelines:

Effect Size	Hedges' g	Observed in This Study
Negligible	< 0.2	–
Small	$0.2 - 0.5$	–
Medium	$0.5 - 0.8$	Lead time ($g = 0.76$)
Large	> 0.8	Alignment (1.31), PAC gap (1.57), Low-PAC targeting (4.47)

The primary results show uniformly large effect sizes, with the low-PAC targeting metric showing a very large effect ($g = 4.47$). These effect sizes substantially exceed the threshold for clinical significance and would be considered strong evidence in biomedical research.

8.3 Multiple Comparison Considerations

Six controllers were compared, and six metrics were computed for each comparison. While formal Bonferroni correction would adjust the significance threshold to $p < 0.05/36 = 0.0014$, all primary comparisons (TCN vs Reactive) yielded $p < 0.001$, which survives even conservative correction. The primary hypothesis was pre-specified (TCN Predictive vs Reactive Threshold on alignment), and all secondary comparisons are consistent in direction and magnitude.

8.4 Reproducibility Details

Component	Specification
Programming language	Python 3.13.3
Deep learning framework	PyTorch (MPS backend, Apple Silicon)
Random seeds	All fixed at 42
Operating system	macOS (Darwin), Apple Silicon
GPU	Apple M-series (MPS), no cloud GPU required
Total codebase	~6,000 lines (production) + archived experiments
Version control	Git, 24+ commits on main branch

Section 9: Conclusions and Future Work

9.1 Summary of Key Findings

- 1. Temporal advantage:** At 5–10 second prediction horizons, the MultiscaleCausalTCN (31,043 parameters) maintains R^2 of 0.24–0.28 while all baselines (persistence, Ridge regression) collapse to negative R^2 . The +0.5 R^2 margin at the operationally relevant range is the TCN's primary contribution.
- 2. Clinical performance on real data:** Validated on all 35 subjects' real EEG recordings, the TCN predictive controller achieved 72.1% alignment vs 64.5% for reactive control (Hedges' $g = 1.31$, $p < 0.001$), with 82.6% vs 51.7% targeting of low-PAC therapeutic windows ($g = 4.47$, $p < 0.001$). This represents a 60% improvement in therapeutic precision.
- 3. Universal benefit:** All 35 out of 35 subjects showed higher clinical utility with TCN-based control (binomial $p < 0.001$), including 6 held-out test subjects never seen during training.
- 4. Near-oracle performance:** The TCN controller's PAC targeting gap (30.5×10) reaches 92% of the theoretical oracle bound (33.3×10), indicating that learned temporal prediction nearly saturates the achievable performance.
- 5. Individual differences validate personalization:** Half of subjects habituate during stimulation while half do not (17/35 vs 18/35). This heterogeneity validates the fundamental need for adaptive, personalized control rather than one-size-fits-all protocols.
- 6. Robust across assumptions:** The adaptive advantage holds across all threshold parameters tested ($\Delta z = 0.2$ – 1.0) and all four fatigue model assumptions (+6.9% to +19.0%, all $p < 0.001$).
- 7. Architecture ceiling:** Eight static prediction architectures spanning 200 to 1.1 million parameters all converged near $R^2 = 0.287$, establishing a data ceiling for single-window PAC prediction from 7 frontal channels.

9.2 Limitations

1. **Offline replay, not live closed-loop.** Validation uses post-hoc replay of recorded EEG. The system makes decisions on real brain data but cannot observe the brain's response to those decisions.
2. **Single dataset.** All results derive from one cohort of 35 Iranian dementia patients. Cross-population generalization is unknown.
3. **Short sessions.** Recording sessions were 6–10 minutes. Clinical protocols run 30–60 minutes, and long-term habituation dynamics may differ.
4. **PAC as the sole biomarker.** PAC is a validated biomarker, but clinical outcomes (amyloid clearance, cognitive improvement) were not measured.
5. **Heuristic controller thresholds.** The z-score thresholds and hysteresis parameters were set by domain knowledge, not optimized via reinforcement learning.
6. **Modest absolute R^2 .** The TCN's R^2 of 0.25 at 5–10 second horizons means 75% of variance is unexplained. The value lies in the +0.5 margin over baselines, not in absolute accuracy.
7. **PAC feature dependence.** Ridge feature ablation showed that without PAC-derived features, R^2 drops from 0.812 to 0.045. The temporal model is substantially autoregressive on PAC history.

9.3 Future Directions

1. **Live closed-loop validation:** Deploy with real-time EEG streaming for prospective crossover validation.
2. **Longer sessions:** Record 30–60 minute sessions to capture the full habituation time course.
3. **Reinforcement learning controller:** Replace heuristic thresholds with an RL agent that learns optimal stimulation policies from data.
4. **Multi-biomarker control:** Extend to multi-biomarker input (PAC + spectral power + connectivity + heart rate).
5. **Cross-site validation:** Test on datasets from different populations and clinical sites.
6. **Embedded deployment:** Both models (1,457 + 31,043 parameters) are small enough for wearable device deployment.

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